

1.6 Electrochemistry

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.6 Electrochemistry
Difficulty	Easy

Time allowed: 20

Score: /11

Percentage: /100

Question 1

Which row correctly describes oxidation and reduction in terms of the transfer of electrons and changes in oxidation state?

	Transfer of electrons		Change in oxidation state	
	Oxidation	Reduction	Oxidation	Reduction
A	Gain	Loss	Increase	Decrease
B	Loss	Gain	Increase	Decrease
C	Loss	Loss	Decrease	Increase
D	Gain	Gain	Decrease	Increase

[1 mark]

Question 2

The heptahydrate mineral of magnesium sulfate is used in Epsom Salts for preparing therapeutic baths. Magnesium sulfate is also used in some medical applications.

What is the oxidation state of MgSO_4 ?

- A. -2
- B. +2
- C. +7
- D. 0

[1 mark]

Question 3

Which row describes the most common oxidation numbers of fluorine, oxygen and hydrogen in compounds?

	F	O	H
A	+7	-1	-1
B	+1	+2	+1
C	-7	-1	-1
D	-1	-2	+1

[1 mark]

Question 4

Solutions containing chlorate(I) ions decompose on heating as shown:



Which row is correct for the oxidation state of the chlorine in each ion?

	ClO^-	ClO_3^-	Cl^-
A	+1	+5	-1
B	+1	+3	+1
C	-1	+5	-1
D	-1	+3	+1

[1 mark]

Question 5

A student added a solution of iodine to a solution of sodium bromide. Which statement correctly describes what happens?

- A. No reaction occurs.
- B. The bromide ions are oxidised.
- C. The iodine ions are oxidised.
- D. Both the bromide and iodide ions undergo changes in their oxidation state.

[1 mark]

Question 6

In which of the following reactions is the change in oxidation number the smallest for nitrogen?

- A. $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$
- B. $3\text{Mg} + \text{N}_2 \rightarrow \text{Mg}_3\text{N}_2$
- C. $4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$
- D. $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$

[1 mark]

Question 7

In the compound $[\text{ICl}_2]^+[\text{SbCl}_6]^-$, the oxidation number of chlorine is -1 .

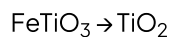
What are the oxidation numbers of I and Sb in the compound?

	I	Sb
A	+1	+5
B	+1	+7
C	+3	+5
D	+3	+7

[1 mark]

Question 8

Titanium dioxide is obtained from the ore ilmenite, Fe(II)TiO_3 .



What is the change in the oxidation number of titanium in the reaction?

- A. +4 to +5
- B. +3 to +4
- C. No change in oxidation number occurs
- D. +6 to +4

[1 mark]

Question 9

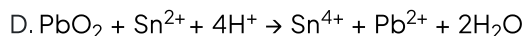
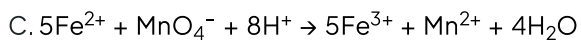
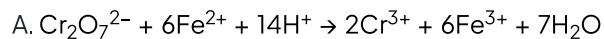
In which of the following compounds is the oxidation state of nitrogen the highest?

- A. NaNO_2
- B. N_2O_4
- C. N_2O
- D. NO

[1 mark]

Question 10

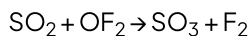
In which reaction does an element have the largest change in oxidation number?



[1 mark]

Question 11

Sulfur dioxide and oxygen difluoride react together. The oxidation number of fluorine in OF_2 is -1.



Which row is correct?

	Sulfur	Fluorine	Oxygen in OF_2
A	Oxidised	Reduced	Reduced
B	Reduced	Oxidised	Reduced
C	Oxidised	Oxidised	Reduced
D	Reduced	Oxidised	Oxidised

[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- In terms of electron transfer, oxidation refers to the **loss of electrons** and reduction refers to the **gain of electrons**: OIL-RIG (oxidation is loss – reduction is gain).
- In terms of oxidation state, the oxidation number **increases** when oxidation has occurred and **decreases** when reduction has occurred.

A, C & D are incorrect as these options do not match the meaning of oxidation and reduction.

2

The correct answer is **D** because:

- In an ionic or covalent compound, the oxidation state of the compound is **zero**.
- For MgSO_4 , the oxidation number of Mg is +2; oxygen is -2 and S is +6.
- Adding them up together gives 0: $(+2) + (+6) + (4 \times -2) = 0$.

A, B & C are incorrect as the oxidation state of MgSO_4 is 0.

3

The correct answer is **D** because:

- Fluorine is the **most electronegative** element, and its oxidation number is always -1.
- The oxidation number of oxygen is always -2 except when in **peroxides** (in this case the oxidation number is -1) or when combined with **fluorine** where it adopts a number to provide an overall value of zero for the compound.

- E.g. in F_2O oxygen is in the +2 state since fluorine is more electronegative and will always have an oxidation number of -1 in compounds, so $(2 \times -1) + 2 = 0$.
- Hydrogen always has oxidation number +1 except when in **metal hydrides** where it is -1.

A, B & C are incorrect as these options do not provide the correct oxidation number of each atom in compounds.

4

The correct answer is **A** because:

- In ClO^- the oxidation number of the chloride ion is +1 since the oxygen is -2 and $1 + (-2) = -1$ which is the charge on the ion.
- In ClO_3^- the oxidation number of the chloride ion is +5 since the oxygen is -2 and $3 \times -2 = -6 + 5 = -1$ which is the charge on the ion.
- In Cl^- the oxidation number is -1 since it is an ion consisting of only one atom.

B, C & D are incorrect as these options do not give the correct oxidation numbers of the ions in the reaction.

5

The correct answer is **A** because:

- The oxidising power **decreases** moving down Group VII.
- Iodine sits below bromine so is unable to oxidise the bromide ion and displace it, so no reaction occurs.

B & C are incorrect as as neither the bromide ions nor iodine atoms are oxidised.

D is incorrect as no reaction occurs, so there are no changes in oxidation states.

6

The correct answer is **D** because:

- The oxidation number of nitrogen in reaction **D** changes from +2 to +4, so an overall change of +2.

A and **B** are incorrect as in both reactions the oxidation number of nitrogen changes from 0 to -3.

C is incorrect as the oxidation number of nitrogen changes from -3 to +2.

7

The correct answer is **C** because:

- Iodine
 - Two chlorines giving a total of -2.
 - One iodine and the ion has a +1 charge so iodine oxidation state must be +3.
- Antimony
 - Six chlorines, giving a total of -6.
 - One antimony and the ion has a charge of -1, so the antimony oxidation state must be +5.

A, B & D are incorrect as these answers do not correspond to the oxidation numbers in the ions.

8

The correct answer is **C** because:

- In ilmenite the oxidation number of titanium is **+4** as each oxygen is -2 and iron is +2, so $2 + 4 + (3 \times -2) = 0$.
- In titanium dioxide the oxidation number of titanium is **+4**, since each oxygen is -2, hence $4 + (2 \times -2) = 0$.
- In both cases the oxidation number is +4, hence the oxidation number doesn't change.

A, B & D are incorrect as there is no change in the oxidation number of titanium.

9

The correct answer is **B** because:

- In N_2O_4 the oxidation state of nitrogen is +4 since each oxygen is -2 and $(4 \times -2) + (2$

The correct answer is **B** because:

- In N_2O_4 the oxidation state of nitrogen is +4 since each oxygen is -2 and $(4 \times -2) + (2 \times +4) = 0$ which is what the oxidation number for N_2O_4 must be as it is a compound.

A is incorrect as in NaNO_2 the oxidation state of nitrogen is +3.

C is incorrect as in N_2O the oxidation state of nitrogen is +1.

D is incorrect as in NO the oxidation state of nitrogen is +2.

10

The correct answer is **C** because:

- In MnO_4^- the oxidation state of Mn is +7 since each oxygen is -2 and $(4 \times -2) + (+7) = -1$ which is what the oxidation number for MnO_4^- must be as it is an ion with a -1 charge.
- The Mn^{2+} ion has an oxidation number of +2.
- The oxidation state of Mn has gone from +7 to +2 which is the largest change in oxidation number.

A is incorrect as the oxidation state of Cr has gone from +6 to +3; the oxidation state of Fe has gone from +2 to +3.

B is incorrect as the oxidation state of Cl has gone from +1 to +5.

D is incorrect as the oxidation state of Pb has gone from +4 to +2; the oxidation state of Sn has gone from +2 to +4.

11

The correct answer is **C** because:

- Sulfur is oxidised as its oxidation number increases from +4 to +6.
- Fluorine is oxidised as its oxidation number increases from -1 to 0.
- Oxygen is reduced as its oxidation number decreases from +2 to -2.

A, B & D are incorrect as these options do not correctly describe the changes in the reaction.